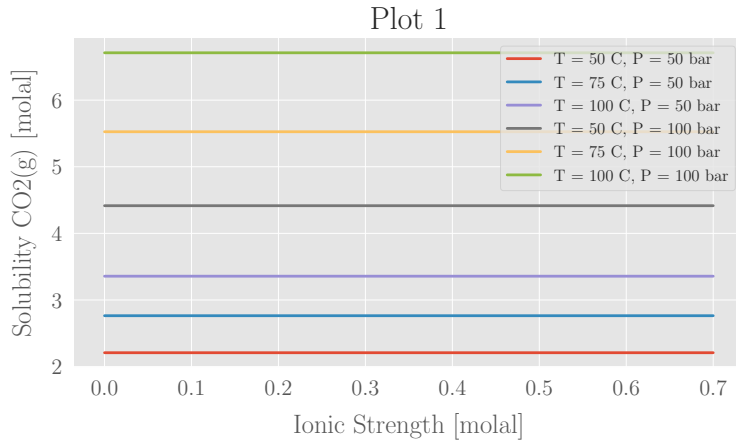
GEOFLUIDS PROJECT: RESULTS

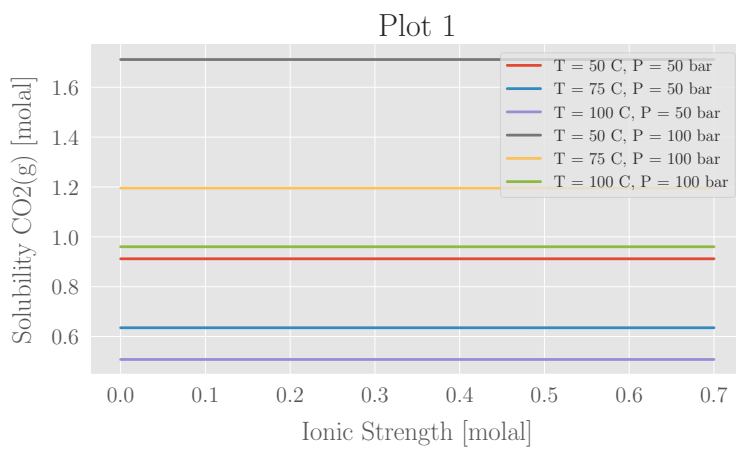
Calculating the solubility of CO_2 and calcite in brine using a Gibbs energy minimization approach

Lecturer: Dr. Svetlana Kyas and Dr. Allan Leal

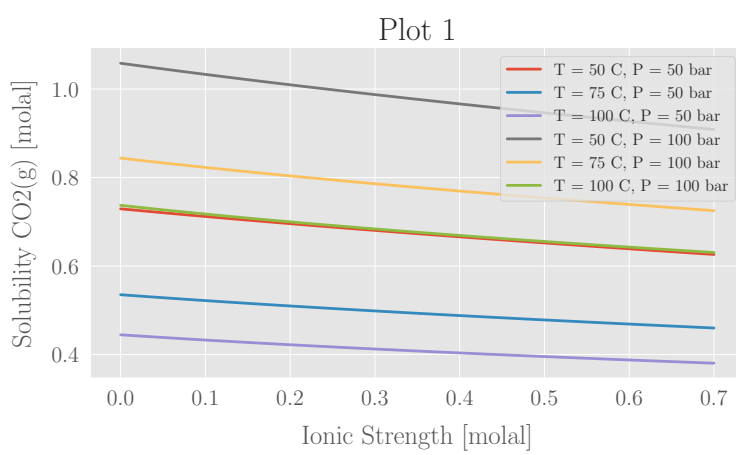
April 23, 2021



(a) Performed **without corrections** on the species activities, a_i, and **without interpolation** for the standard chemical potentials, μ_i°.

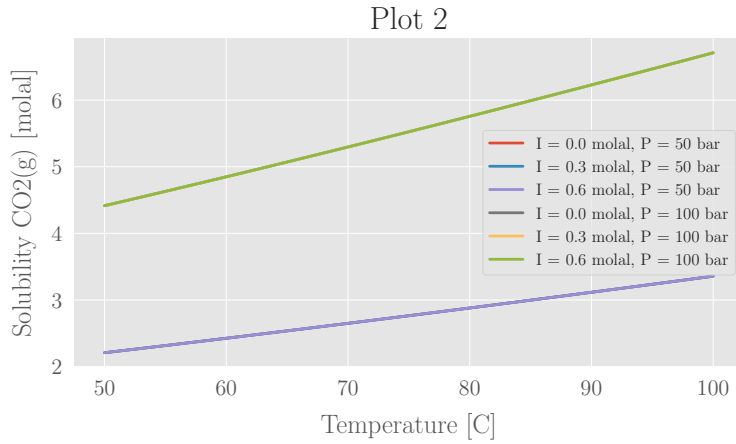


(b) Performed **without corrections** on the species activities, a_i, but **with interpolation** for the standard chemical potentials, μ_i°.

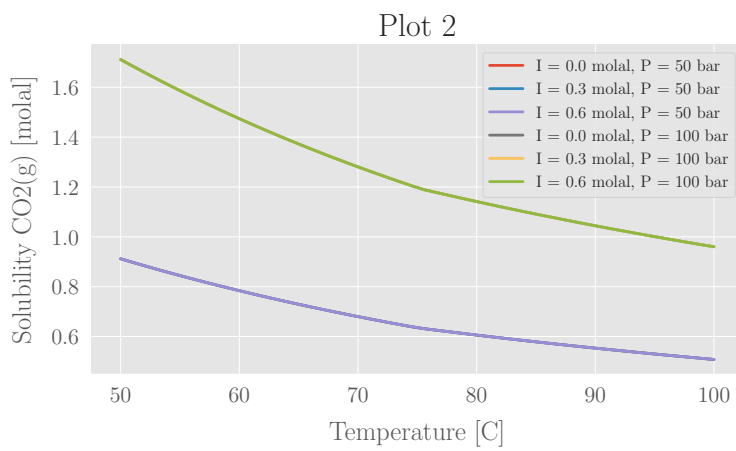


(c) Performed **with corrections** on the species activities, a_i, and **with interpolation** for the standard chemical potentials, μ_i°.

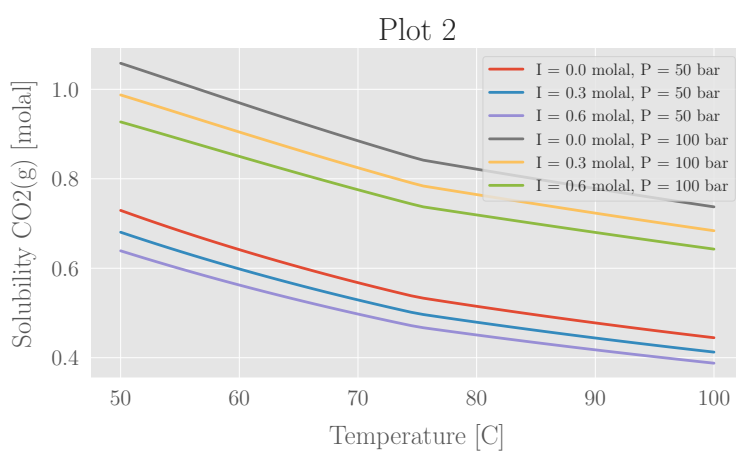
Figure 1: Plot 1 performed for three different scenarios.



(a) Performed **without corrections** on the species activities, a_i, and **without interpolation** for the standard chemical potentials, μ_i°.

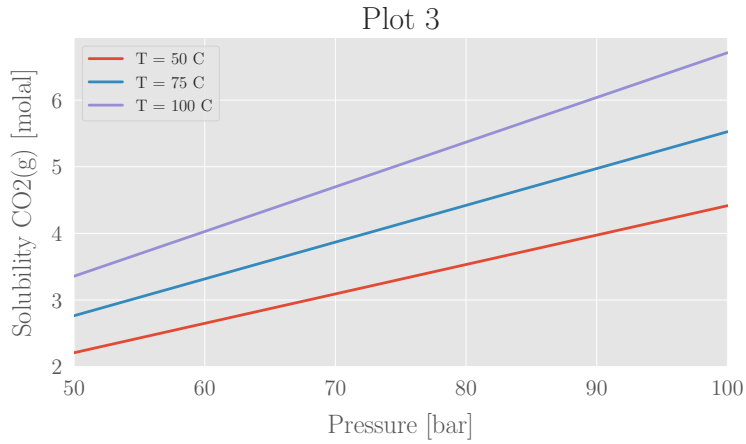


(b) Performed **without corrections** on the species activities, a_i, but **with interpolation** for the standard chemical potentials, μ_i°.

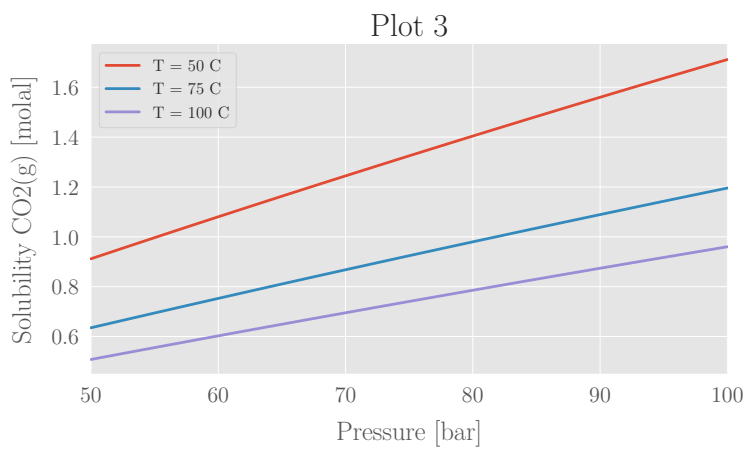


(c) Performed **with corrections** on the species activities, a_i, and **with interpolation** for the standard chemical potentials, μ_i°.

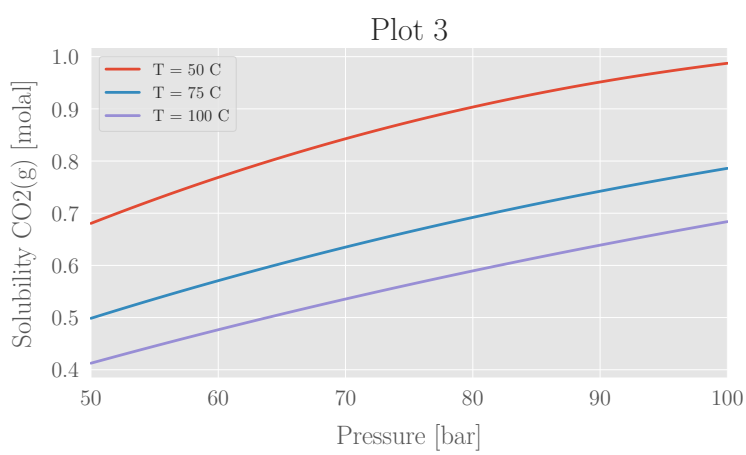
Figure 2: Plot 2 performed for three different scenarios.



(a) Performed **without corrections** on the species activities, a_i, and **without interpolation** for the standard chemical potentials, μ_i°.

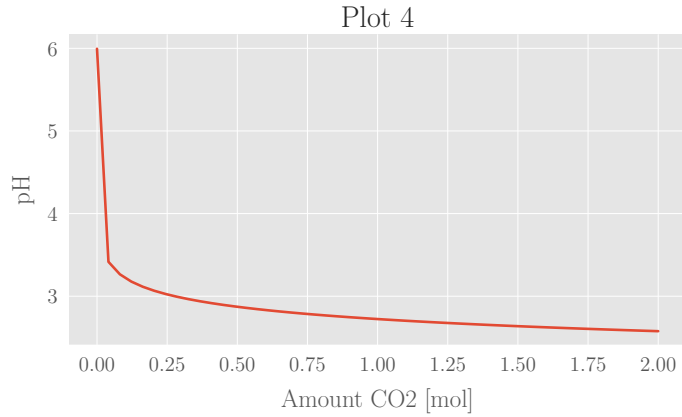


(b) Performed **without corrections** on the species activities, a_i, but **with interpolation** for the standard chemical potentials, μ_i°.

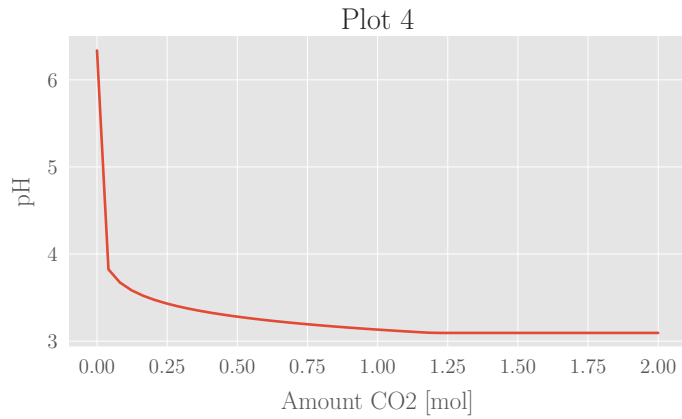


(c) Performed **with corrections** on the species activities, a_i, and **with interpolation** for the standard chemical potentials, μ_i°.

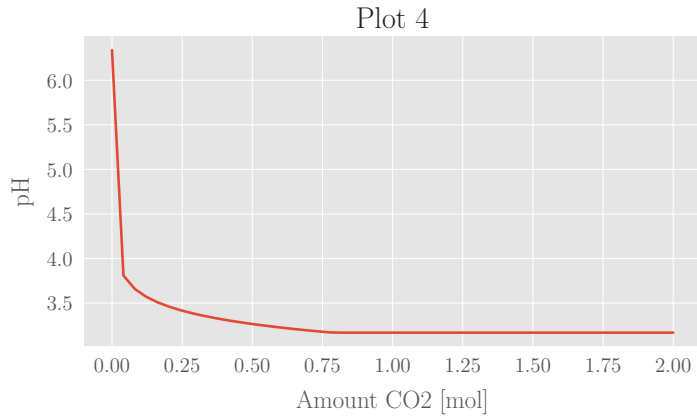
Figure 3: Plot 3 performed for three different scenarios.



(a) Performed **without corrections** on the species activities, a_i, and **without interpolation** for the standard chemical potentials, μ_i°.

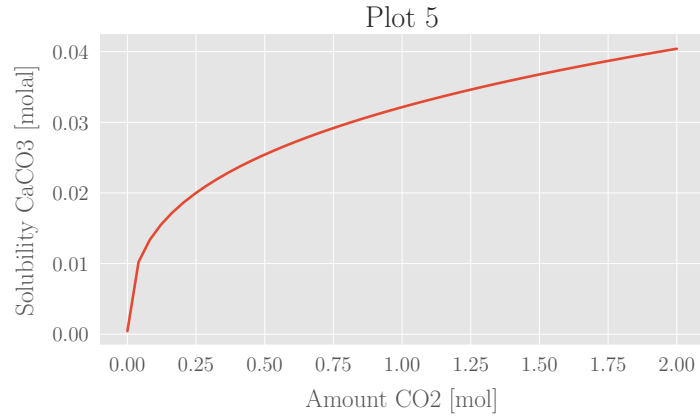


(b) Performed **without corrections** on the species activities, a_i, but **with interpolation** for the standard chemical potentials, μ_i°.

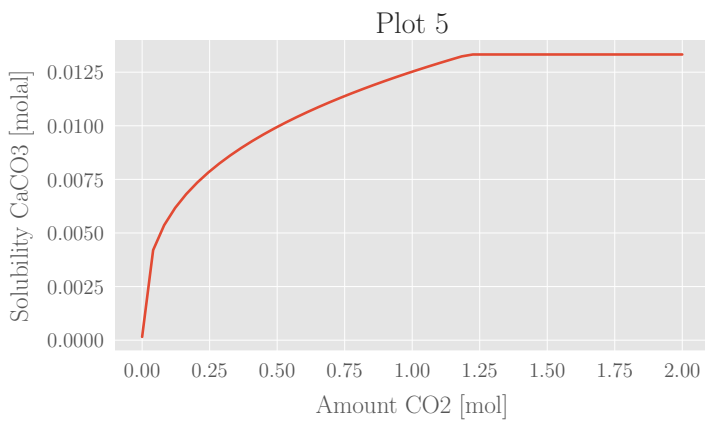


(c) Performed **with corrections** on the species activities, a_i, and **with interpolation** for the standard chemical potentials, μ_i°.

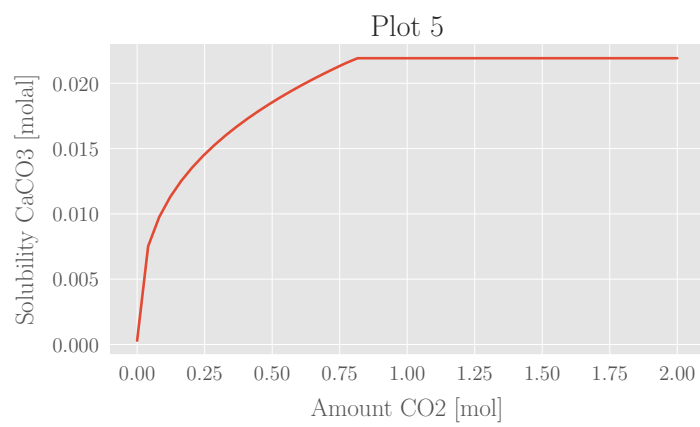
Figure 4: Plot 4 performed for three different scenarios.



(a) Performed **without corrections** on the species activities, a_i, and **without interpolation** for the standard chemical potentials, μ_i°.

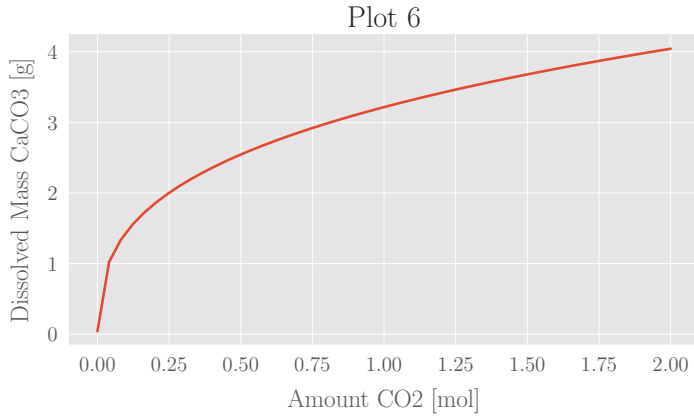


(b) Performed **without corrections** on the species activities, a_i, but **with interpolation** for the standard chemical potentials, μ_i°.

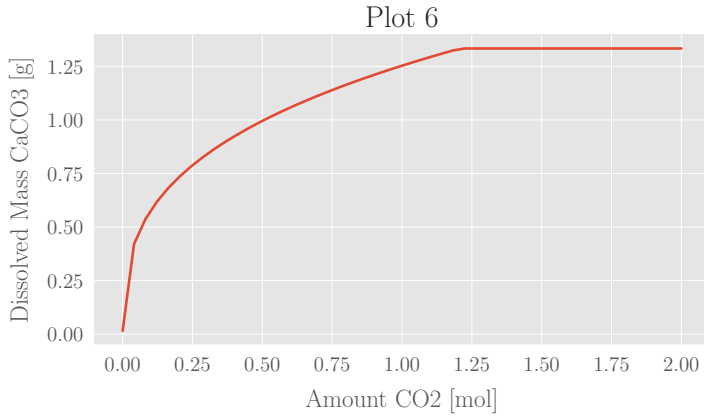


(c) Performed **with corrections** on the species activities, a_i, and **with interpolation** for the standard chemical potentials, μ_i°.

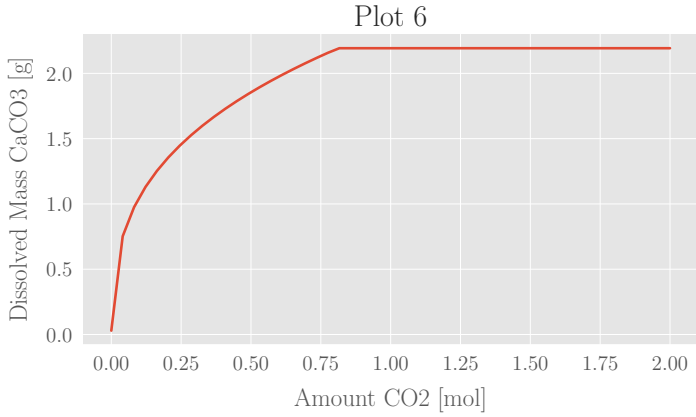
Figure 5: Plot 5 performed for three different scenarios.



(a) Performed **without corrections** on the species activities, a_i, and **without interpolation** for the standard chemical potentials, μ_i°.

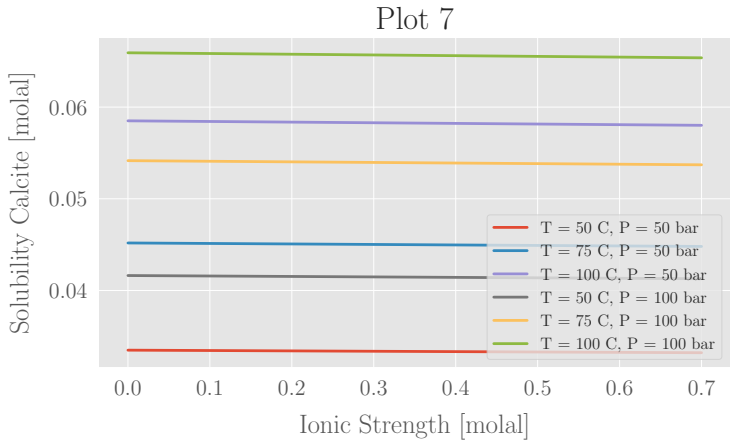


(b) Performed **without corrections** on the species activities, a_i, but **with interpolation** for the standard chemical potentials, μ_i°.

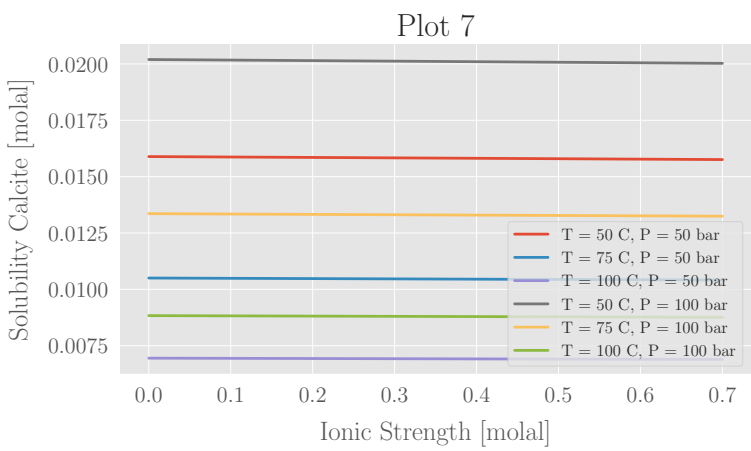


(c) Performed **with corrections** on the species activities, a_i, and **with interpolation** for the standard chemical potentials, μ_i°.

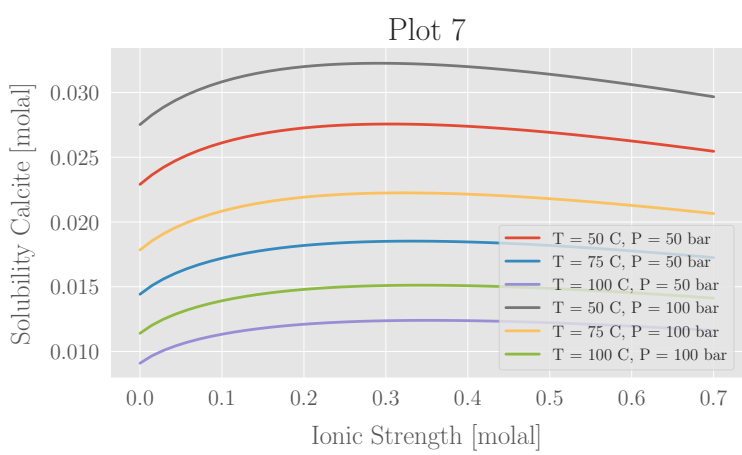
Figure 6: Plot 6 performed for three different scenarios.



(a) Performed **without corrections** on the species activities, a_i, and **without interpolation** for the standard chemical potentials, μ_i°.

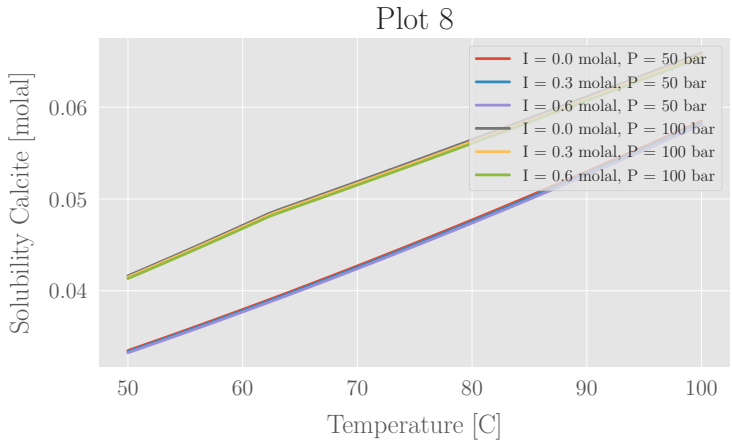


(b) Performed **without corrections** on the species activities, a_i, but **with interpolation** for the standard chemical potentials, μ_i°.

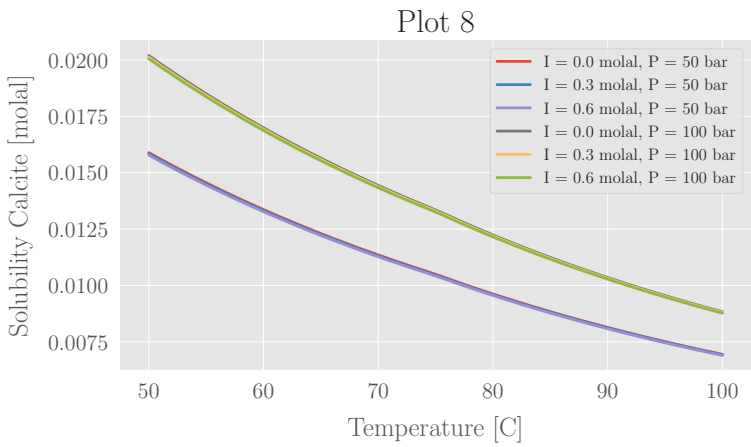


(c) Performed **with corrections** on the species activities, a_i, and **with interpolation** for the standard chemical potentials, μ_i°.

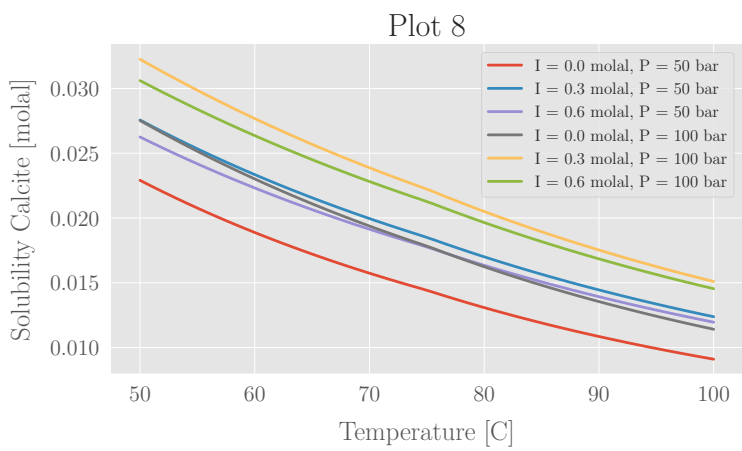
Figure 7: Plot 7 performed for three different scenarios.



(a) Performed **without corrections** on the species activities, a_i, and **without interpolation** for the standard chemical potentials, μ_i°.

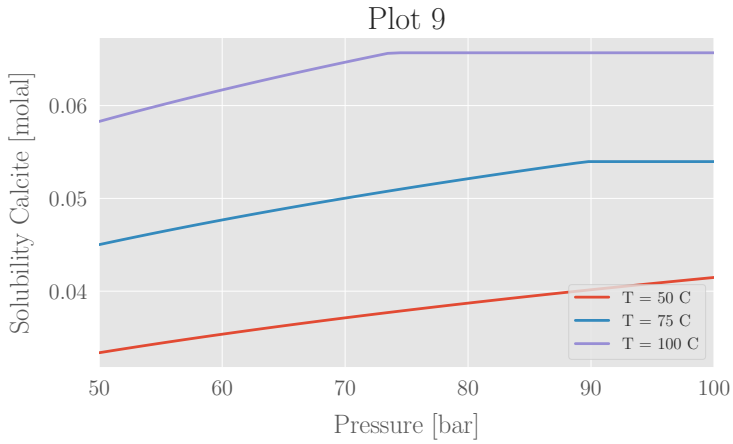


(b) Performed **without corrections** on the species activities, a_i, but **with interpolation** for the standard chemical potentials, μ_i°.

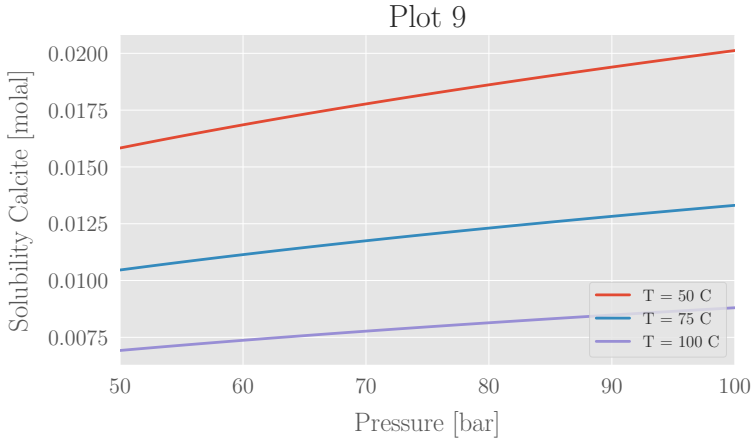


(c) Performed **with corrections** on the species activities, a_i, and **with interpolation** for the standard chemical potentials, μ_i°.

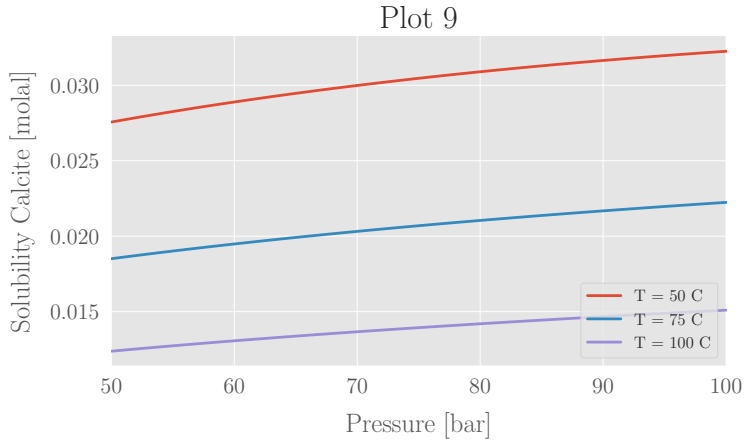
Figure 8: Plot 8 performed for three different scenarios.



(a) Performed **without corrections** on the species activities, a_i, and **without interpolation** for the standard chemical potentials, μ_i°.



(b) Performed **without corrections** on the species activities, a_i, but **with interpolation** for the standard chemical potentials, μ_i°.



(c) Performed **with corrections** on the species activities, a_i, and **with interpolation** for the standard chemical potentials, μ_i°.

Figure 9: Plot 9 performed for three different scenarios.